NAT and PAT Concepts

1. Theory

1.1 What is NAT?

Network Address Translation (NAT) is a technique performed by a router (or firewall) that rewrites the IP address information in packet headers as they pass between two networks. In practice it lets devices that use private IPv4 addresses (RFC 1918: 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16) communicate with the public Internet, which only routes public addresses. The router keeps a NAT translation table so it can map replies back to the right internal device.

1.2 What is PAT?

Port Address Translation (PAT), also called NAT overload, is a form of dynamic NAT in which many inside hosts share one public IP address. The router tells each conversation apart by its Layer 4 port number: it rewrites the source IP and the source port, and records the pair in the translation table. Because there are about 65,000 ports per IP address, a single public address can carry thousands of simultaneous sessions.

1.3 Types of NAT

Type	Mapping	Typical use
Static NAT	One inside address to one fixed public address (1:1)	Servers that must be reachable from outside, such as a web or mail server
Dynamic NAT	Inside addresses to a pool of public addresses (many:pool)	Small networks that own several public IPs
PAT (overload)	Many inside addresses to one public address, separated by port numbers	Homes, offices, and ISPs; the most common form

1.4 NAT Terminology

Term	Meaning
Inside local	Private address of the inside host as seen on the inside network (e.g. 192.168.1.10)
Inside global	Public address that represents the inside host to the outside world (e.g. 203.0.113.5)
Outside local	Address of the outside host as seen from the inside network
Outside global	Actual address of the outside host on the outside network (e.g. 203.0.113.10)

1.5 Importance: IPv4 Address Conservation

IPv4 has only about 4.3 billion addresses, far fewer than the number of phones, laptops and IoT devices in use. NAT and PAT stretch this limited supply:

- **Address conservation:** an entire home, campus or mobile network can sit behind one or a few public IPs.
- **Cost saving:** organisations lease far fewer public addresses from their ISP.
- **Security by obscurity:** internal addressing is hidden, and unsolicited inbound traffic has no translation entry, so it is dropped by default.
- **Flexibility:** the internal network can be renumbered or the ISP changed without touching every host.

Limitation: NAT breaks true end-to-end connectivity and can complicate protocols that embed IP addresses in their payload. IPv6 removes the need for NAT, but IPv4 plus PAT remains dominant today.

2. Practical: Topology and Addressing

Devices used in Cisco Packet Tracer: one router (Cisco 2911 or 1941), one switch (2960, from Task 2 onward), up to three inside PCs, and one external PC that also acts as the public web server (enable its HTTP service).

```
PC0 (192.168.1.10) --+
PC1 (192.168.1.11) --+-- SW1 --- Gi0/0 [ R1 ] Gi0/1 --- Ext-PC / Web Server
PC2 (192.168.1.12) --+   (Task 4)  192.168.1.1  203.0.113.1    203.0.113.10

<------- INSIDE (private) ------->            <---- OUTSIDE (public) ---->
```

Device	Interface	IP Address	Subnet Mask	Gateway
R1	Gi0/0 (inside)	192.168.1.1	255.255.255.0	n/a
R1	Gi0/1 (outside)	203.0.113.1	255.255.255.0	n/a
PC0	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.1
PC2 (Task 4)	FastEthernet0	192.168.1.12	255.255.255.0	192.168.1.1
Ext-PC / Web Server	FastEthernet0	203.0.113.10	255.255.255.0	203.0.113.1

Note: 203.0.113.0/24 is a documentation range, used here to stand in for "public" addresses. Any public-looking range works in the simulator.

[Insert screenshot: final topology in Packet Tracer with all devices labelled]

3. Task 1: Static NAT

Objective: build the network with one internal PC and one external PC, then configure static NAT so the internal PC (192.168.1.10) is always represented by the public address 203.0.113.5.

Steps

1. Place R1, PC0 and Ext-PC. Connect PC0 to R1 Gi0/0 and Ext-PC to R1 Gi0/1 with copper cross-over cables.

2. Assign the IP addresses and gateways from the addressing table to PC0 and Ext-PC (Desktop > IP Configuration).
3. Configure the router interfaces and mark the inside and outside sides:

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# ip nat inside
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 203.0.113.1 255.255.255.0
R1(config-if)# ip nat outside
R1(config-if)# no shutdown
R1(config-if)# exit
```

4. Create the static mapping (inside local to inside global):

```
R1(config)# ip nat inside source static 192.168.1.10 203.0.113.5
R1(config)# end
R1# copy running-config startup-config
```

5. Verify with show running-config and show ip nat translations (the static entry appears immediately, before any traffic).

```
R1# show ip nat translations
Pro  Inside global   Inside local    Outside local   Outside global
---  203.0.113.5     192.168.1.10    ---             ---
```

[Insert screenshot: R1 configuration and static NAT entry]

4. Task 2: Port Address Translation (PAT)

Objective: let two internal PCs (PC0 and PC1) reach the simulated public web server using one public IP address, which is the router's outside interface (203.0.113.1).

Steps

1. Remove the static mapping and clear old entries so the two tasks do not interfere:

```
R1(config)# no ip nat inside source static 192.168.1.10 203.0.113.5
R1(config)# end
R1# clear ip nat translation *
```

2. Add a switch SW1: connect R1 Gi0/0 to SW1, then connect PC0 and PC1 to SW1. Give PC1 the address 192.168.1.11/24 with gateway 192.168.1.1.
3. On Ext-PC (server), open Services > HTTP and make sure HTTP is On, so it behaves as the public web server.
4. Define which inside addresses may be translated, then enable overload:

```
R1# configure terminal
R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)# ip nat inside source list 1 interface gigabitEthernet 0/1
overload
R1(config)# end
R1# copy running-config startup-config
```

How it works: access-list 1 selects the inside hosts; the keyword `interface gigabitEthernet 0/1` says to use that interface's public IP as the inside global address; `overload` enables PAT so many hosts share it, separated by port numbers.

[Insert screenshot: PAT configuration (show running-config | include nat)]

5. Task 3: Testing and Capturing the NAT Table

1. Clear any old entries: R1# clear ip nat translation *
2. On PC0, open Command Prompt and run: ping 203.0.113.10 (expect replies; the first packet may time out while ARP completes).
3. On R1, immediately view the translations before they time out:

```
R1# show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.1:1      192.168.1.10:1     203.0.113.10:1     203.0.113.10:1
icmp 203.0.113.1:2      192.168.1.10:2     203.0.113.10:2     203.0.113.10:2
```

4. Repeat from PC1, and also open a web browser on PC0 and PC1 and visit http://203.0.113.10:

```
R1# show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
tcp  203.0.113.1:1025   192.168.1.10:1025  203.0.113.10:80    203.0.113.10:80
tcp  203.0.113.1:1026   192.168.1.11:1025  203.0.113.10:80    203.0.113.10:80
```

5. Check counters: R1# show ip nat statistics

Note: the port numbers above are sample values. Packet Tracer will show its own numbers; what matters is that every inside local address maps to the same inside global IP (203.0.113.1) with a different port.

Interpreting the result

- Inside local = real private address and port of the PC.
- Inside global = the shared public address and the unique port chosen by the router.
- Outside local/global = the web server's address and port 80 (or the ICMP identifier for ping).

[Insert screenshot: successful ping from PC0 to 203.0.113.10]

[Insert screenshot: output of show ip nat translations]

6. Task 4: Adding a Third Internal PC

1. Add PC2 and connect it to SW1. Set 192.168.1.12/24, gateway 192.168.1.1.
2. Update the NAT configuration. Because the ACL already permits the whole 192.168.1.0/24 network, no change is needed for PC2. If your ACL lists individual hosts, add the new one:

```
R1(config)# access-list 1 permit host 192.168.1.12
```

3. Ping 203.0.113.10 and browse to http://203.0.113.10 from all three PCs, then check the table:

```
R1# show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
tcp  203.0.113.1:1025   192.168.1.10:1025  203.0.113.10:80    203.0.113.10:80
tcp  203.0.113.1:1026   192.168.1.11:1025  203.0.113.10:80    203.0.113.10:80
tcp  203.0.113.1:1027   192.168.1.12:1025  203.0.113.10:80    203.0.113.10:80
```

Result: three different inside local addresses share the single inside global address 203.0.113.1, each with its own port. This is PAT working as intended.

[Insert screenshot: NAT table showing translations for all three PCs]

7. Task 5: NAT/PAT and Large Apps like Zomato or Flipkart

7.1 Explanation in my own words

There are not enough public IPv4 addresses for every phone and laptop that opens Zomato or Flipkart. Instead of giving each device its own public address, home routers and mobile operators use PAT. Thousands of customers sit behind one public IP, and the router keeps a table of which port belongs to which customer. When you open the app, your request leaves with the shared public IP and a unique port; when Flipkart's reply comes back to that IP and port, the router looks it up and forwards it to your phone. That is why millions of users can shop at the same time without millions of public addresses.

On the company side, Zomato or Flipkart need only a small number of public addresses, typically on load balancers and gateways. Their many application and database servers use private addresses inside the cloud network, and a NAT gateway lets those servers reach outside services (for example payment or SMS providers) through a few public IPs. This saves addresses, reduces cost, and keeps internal servers hidden from direct Internet access.

7.2 Two scenarios where a wrong NAT/PAT setup breaks connectivity

Scenario 1: Missing or wrong inside/outside marking or ACL. A restaurant adds a new Wi-Fi subnet for its order tablets, but the NAT access list still permits only the old subnet (or ip nat inside / ip nat outside is applied to the wrong interface). The router refuses to translate the new subnet's traffic, so the tablets cannot reach the delivery platform. Orders stop arriving even though the local network looks healthy and the Internet works for other devices.

Scenario 2: PAT port exhaustion or very short translation timeouts. During a big sale or dinner rush, a shared public IP can run out of free ports, or entries expire too quickly. New connections then fail and long-lived ones drop, so customers see loading spinners, payment pages that time out, or live order tracking that disconnects. The fix is to add more public IPs to the PAT pool, tune the timeouts, or use a larger NAT gateway.

Other common misconfigurations: a static NAT entry that points to the wrong inside server (the public service becomes unreachable), or two inside hosts mapped to the same inside global address (conflicting translations).

8. Conclusion

In this lab I configured static NAT to give an internal PC a fixed public identity, and PAT (NAT overload) to let several internal PCs share one public IP address to reach a public web server. The show ip nat translations output confirmed the mappings. NAT and PAT are essential for conserving IPv4 addresses and are used at every scale, from a home router to the networks of large internet companies.

Quick Command Reference

Command	Purpose
`ip nat inside / ip nat outside`	Mark an interface as the inside or outside NAT interface
`ip nat inside source static <local> <global>`	Create a static NAT mapping
`access-list 1 permit 192.168.1.0 0.0.0.255`	Select inside addresses eligible for translation
`ip nat inside source list 1 interface g0/1 overload`	Enable PAT using the outside interface address
`show ip nat translations`	Display the NAT translation table
`show ip nat statistics`	Display NAT counters and configuration summary
`clear ip nat translation *`	Clear dynamic translation entries
`debug ip nat`	Watch translations in real time (turn off with undebug all)